

See Beyond the Source Code.

DND Source Lens

An intelligent codebase analysis and visualization platform that transforms complex source code into an interactive map of architecture, files, modules, and dependencies.

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Problem Statement: Software and Developer Tools

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Problem Statement

Understanding an Unfamiliar Codebase Is Difficult

Modern software projects often contain:

1. Hundreds or thousands of files
2. Complex dependencies
3. Multiple modules and services
4. Incomplete or outdated documentation
5. Hidden relationships between components

Developers spend significant time manually:

1. Exploring files
2. Searching code
3. Following imports
4. Reading documentation

When a developer joins an existing project, they often ask:

Where does the application start?

Which files are connected?

Where is authentication handled?

How does data move through the system?

What happens when an API request is received?

What if a codebase could automatically explain its own architecture?

Target Users & Why The Problem Matters?

Who needs DND Source Lens?

Developers

Quickly understand unfamiliar projects

Students

Learn how real-world software systems are structured.

Development Teams

Improve onboarding and collaboration.

Open-Source Contributors

Understand large repositories before contributing.

Understanding a codebase is often one of the biggest barriers to:

1. Joining a project
2. Fixing bugs
3. Adding new features
4. Reviewing code
5. Maintaining legacy systems

Existing Solutions / Current Approach

How Developers Currently Understand Codebases

Documentation

Problem: Often incomplete, can become outdated & requires manual maintenance.

IDE Navigation

Developers use: Go to Definition, find References & global Search.

Problem: These tools help developers navigate individual pieces of code but don't always provide a clear high-level architectural view.

Static Analysis Tools

They can provide: Metrics, warnings & Dependency information.

Problem: The output can be technical and fragmented, making it harder to understand the overall system structure.

The opportunity is to transform scattered source code relationships into an understandable, navigable architecture view.

Your Proposed Solution

Introducing DND Source Lens

An Intelligent Lens into Your Codebase.

DND SourceLens automatically analyzes a software project and transforms it into an interactive architecture map.

Key Features:

- 1.Interactive Code Map:** Visualize files, modules, and their relationships.
- 2.Dependency Explorer:** Understand which components depend on each other.
- 3.Smart File Explorer:** Explore the structure and importance of project files.
- 4.Code Intelligence:** Identify functions, classes, imports, and modules.
- 5.Structural Insights:** Detect patterns such as circular dependencies.
- 6.AI-Powered Explanation:** Ask questions and receive understandable explanations about the architecture.
- 7.Tool and Extension:** DND Source Lens is a tool and a Extension too.

How It Works / User Flow

From Source Code to System Understanding

Upload Project

Code Scanner

Code Parser

Dependency Engine Finds Connections

Architecture Graph Interactive Map

Explore + Ask AI



What Makes DND Source Lens Different?

Beyond Code Search. Beyond Static Documentation.

- 1.Codebase as a Navigable Map
- 2.Multi-Level Understanding
- 3.Architecture Navigation
- 4Automatic Relationship Discovery
- 5.AI-Powered Architecture Understanding

The Core Difference

Existing Approach

CODE → SEARCH → READ → SEARCH → READ → UNDERSTAND

DND SourceLens

CODE → ANALYZE → VISUALIZE → EXPLORE → UNDERSTAND

NOTE

DND Source Lens provides both:

- 1.High-level architecture
- 2.Low-level code relationships

The system automatically discovers:

- 1.Imports
- 2.Dependencies
- 3.Module relationships
- 4.Circular dependencies

Instead of manually searching code, developers can ask:

- 1.How does authentication work?
- 2.Which components interact with the database?
- 3.What depends on this module?

Basic Implementation Approach Technology & System Architecture

